

Appendix V

Detailed Thematic Analysis of the Perceived Weaknesses of LitTraceAI

V.0.1 Constraints in User Interaction

This theme reflects expert perceptions regarding the limited support for interactive, flexible, and verifiable engagement with the generated literature review. Importantly, experts referred to different levels of interaction with LitTraceAI. Some interaction constraints were associated with iterative scholarly collaboration and the ability to refine the generated content through human expertise, while others were connected to verification practices and trust in the generated output. Additional interaction concerns focused on the ability to inspect specific aspects of the literature, such as gaps, limitations, or results, as well as interface usability during reading. Although some issues were mentioned by only a small number of experts, they remain analytically-relevant because they introduce important reflections regarding trust, scholarly writing practices, and expectations surrounding AI-assisted literature review generation.

Four out of fourteen experts highlighted limitations in continuing the interaction after the generation process. Experts appeared to expect a more collaborative interaction model

in which they could iteratively refine, question, and improve the generated literature review based on their own expertise and research intentions. Expert 7 explained:

The program [LitTraceAI] does not suggest ideas, unlike ChatGPT, which always invites you to continue the conversation. Here, after giving you a result, the interaction does not continue, it is as if it cuts you off. But what if you wanted to keep improving the content or ask the tool [LitTraceAI] why it did that?

Similarly, another expert stated:

How do you input the information into the tool? Because I gave you the papers, but the idea is that I should interact directly with the tool. (Expert 21)

Another expert suggested that interaction with LitTraceAI could support idea development and writing quality, noting that “*if I start interacting with it, it can help me generate stronger ideas or texts with greater coherence and cohesion*” (Expert 22). Together, these perspectives may indicate that experts perceive literature review generation as an iterative scholarly activity in which human expertise remains important for refining, adjusting, and shaping the final output rather than simply receiving a static generated response. This expectation aligns with an **AI-in-the-Loop** perspective

One out of fourteen experts additionally referred to the possibility of generating different stylistic variations of the literature review. Expert 16 explained:

There might be different stylistic variations in the output. This one gives me a certain style, but what if I want something more concise or with more details.

Although this suggestion appeared with low frequency, it raises analytically relevant questions regarding what constitutes an appropriate writing style for AI-generated literature reviews. The reflection may indicate that experts do not necessarily expect a single universally appropriate style, but rather multiple valid stylistic approaches depending on disciplinary conventions, writing objectives, level of detail, or intended audience. While the

primary goal of LitTraceAI was not to support multiple writing styles, this observation introduces broader questions deserving further research regarding how AI systems could adapt scholarly writing to different academic contexts and user preferences.

Some interaction constraints were more strongly associated with verification and trust in the generated content. One out of fourteen experts suggested that *“the citations should provide access to the PDFs”* (Expert 18). In a follow up conversation, the same expert explained that this functionality was important because, based on previous experiences using systems such as ChatGPT or Perplexity AI, citation links sometimes redirect to broken pages or to papers different from the ones actually cited. This reflection may indicate that direct access to the referenced PDF is associated not only with convenience, but also with verification practices and confidence in the reliability of AI-generated references.

Similarly, one out of fourteen experts suggested that LitTraceAI should highlight directly the source passages used to generate the sentences:

It [LitTraceAI] should highlight the texts on which the sentences are based in the original papers. The tool shows me the sentences that are supposedly in the papers, but who assures me that the AI did not invent those sentences? (Expert 20)

In a follow up explanation, Expert 20 emphasised that highlighting *“the sentences used to construct the text”* directly within the original paper was *“the only way”* to trust that LitTraceAI was not *“inventing the sentences.”* The expert further associated this functionality with the importance of *“verification,”* suggesting that merely displaying extracted source text may not always be sufficient for establishing confidence in the relationship between the generated content and the original literature. Although mentioned by a single expert, this perspective contributes a deeper level of analysis regarding how traceability mechanisms may shape trust in AI-generated academic writing.

Other experts focused on interaction features intended to support the inspection of specific elements of the literature review. One out of fourteen experts noted that the tool

“does not allow me to filter the sentences I am interested in, for example gaps, limitations, or results” (Expert 23). Although this concern appeared infrequently, it introduces an analytically-relevant perspective regarding how users may expect AI systems to support selective exploration of the literature according to specific analytical interests and review objectives.

Finally, some experts referred to interaction improvements associated with provenance visibility and source identification during reading. Four out of fourteen experts described difficulties maintaining access to provenance information while navigating the generated text. Expert 19 explained:

You cannot see the reference information or the provenance as you move down the section. For example, in the last section I always have to go back up to see the citation or the provenance.

Another expert suggested that the provenance interface *“should be more user friendly ... so that I do not have to go to the top of the screen to see the provenance details”* (Expert 20). Additionally, Expert 21 noted that the provenance should include *“not only the title of the original paper but also the authors so that I can identify the paper more quickly because usually one does not remember the title.”*

These reflections may indicate opportunities to improve navigation and source recognition during reading. However, experts appeared to frame these aspects more as usability and interaction enhancements rather than as limitations fundamentally affecting engagement with the generated content itself.

V.0.2 Constrained Engagement with Source Material

This theme reflects expert perceptions regarding the limited way in which LitTraceAI engages with the provided literature. Experts referred to constraints affecting different levels of engagement with source material, including how the tool accesses papers, compares and

integrates ideas across studies, relies on interpretations produced by other authors, and generates inferences beyond explicitly stated content. While some concerns focused on the ability of LitTraceAI to synthesise relationships between concepts, others questioned the extent to which the tool could move beyond the information directly available in the source material. Although certain observations were mentioned by only a small number of experts, they remain analytically relevant because they introduce important considerations regarding scholarly interpretation, contextualization, and knowledge synthesis in AI-generated literature reviews.

Three out of eight experts highlighted limitations in how LitTraceAI compared and connected ideas across studies. Expert 6 explained:

In topics where the tool [LitTraceAI] has to contrast considerably, the ideas that the tool connects are not entirely clear. Therefore, the expert is needed to refine the information.

The same expert noted additionally that, when the literature is less-clearly defined or conceptually fragmented, “*the AI struggles to contrast,*” giving as an example the relationship between “*agglomeration and informality,*” where “*it is not clear why the AI connected these two topics.*” This observation may indicate that LitTraceAI experiences difficulties articulating conceptual relationships when the literature contains ambiguous or weakly defined connections.

Similarly, Expert 24 questioned the limited scope of variables and perspectives considered by LitTraceAI:

It limitedly chooses topics such as in the case of organisational identification as the only variable, when there are many more complex variables that explain UPB.

The expert further explained that LitTraceAI focused on “*MES*” despite it “*not being the only scale for measuring ethics.*” Together, these perspectives may indicate that the

generated synthesis can sometimes privilege narrower conceptual interpretations, limiting the integration of alternative theoretical perspectives or explanatory variables across studies.

Two out of eight experts also referred to constraints associated with how LitTraceAI accesses source material. Expert 21 explained:

A limitation is that it does not process the full paper. In my field, the discussion section typically includes analysis and insights that are central to the research.

This observation reflects the limitation of **reliance on background sections** described in Section 7.1.2, where outputs depend on information extracted from selected sections of papers and may therefore exclude relevant information reported elsewhere. This perspective may indicate that relevant information for constructing a literature review is not always confined to literature review or introduction sections. Depending on the research area, important interpretations and contextual insights may also be reported in discussion or analytical sections related to the research topic. However, as this limitation was mentioned by only two experts, there is limited evidence that it represents a general limitation of LitTraceAI within the studied sample.

Additionally, two out of eight experts raised concerns regarding LitTraceAI's dependence on interpretations already reported by other authors rather than direct engagement with the original source material. Expert 8 explained:

It [generated sentence] is not based on the source text. It [generated sentence] is based on what one author said about another author. This is sometimes dangerous because, for example, students assume that what that author says about the original author is correct, and in many cases it is incorrect or incomplete.

Relatedly, while inspecting the provenance information associated with the generated text, Expert 15 observed that *“in the first section, it [LitTraceAI] cites a study to identify harm potentials where an evaluation has been conducted, but does not cite the original paper.”* The expert further described the output as *“taking the content of one of the papers”*

and “*copying it.*” Similarly, during the analysis of the provenance information, Expert 21 questioned:

Am I treating the AI [LitTraceAI] as if it were an author? Then, why did the AI [LitTraceAI] not read the original author?

These observations may indicate expectations that AI-generated literature reviews should engage more directly with primary sources rather than relying on interpretations mediated by other authors. However, these comments should also be understood in relation to the design goals of LitTraceAI. The tool was primarily oriented toward verifiability rather than factuality, meaning that it provides mechanisms allowing users to inspect and verify the provenance of generated content rather than guaranteeing that all references correspond directly to the original source author.

In this sense, the experts’ observations may not necessarily indicate a system error, but rather a tension between expert expectations of direct engagement with primary sources and the tool’s emphasis on supporting verification practices through provenance inspection.

Finally, one out of eight experts referred to limitations in generating inferences beyond the explicit content of the literature. Expert 1 stated that “*it does not infer because inferring is part of the research process that we do as future work*” (Expert 1). Although this observation appeared infrequently, it introduces an analytically-relevant perspective regarding the extent to which AI-generated literature reviews should generate broader interpretations or future research directions beyond the explicit content of the provided sources.

However, this perspective should also be understood in relation to the broader role assigned to the tool by other experts. As discussed in Section ??, some experts perceived the AI primarily as a cognitive support resource for synthesising, organising, and comparing literature rather than as a system intended to fully replace the researcher’s interpretive role. In this sense, this observation may introduce a broader question regarding AI-assisted literature review: **should AI systems generate inferential interpretations for the researcher, or should this remain primarily part of the researcher’s scholarly work?**

V.0.3 Constraints in Academic Integrity

This theme reflects expert concerns regarding plagiarism risks associated with different stages of the generation process. Experts referred not only to the limited transformation of source material in the generated text, but also to the possibility that problematic content originating from the cited papers themselves could be reproduced through the tool. Although discussed by a small number of experts, these observations introduce analytically-relevant considerations regarding originality in AI-assisted literature review generation.

Two out of three experts highlighted concerns regarding the low degree of textual transformation in the generated literature review. Expert 4 explained:

It gives me the impression that there is too much ‘copy paste’ from other people’s material. The risk is that, for example, if this is passed through a plagiarism detector, it may indicate that it was copied and pasted from other sources. In other words, the style ends up being very similar to that of the papers that already dominate the topic. The text should have a little more variation.

Similarly, Expert 27 questioned the practical usability of the generated content because:

It’s not really producing something that’s fully usable, because there’s so little paraphrasing and it’s mostly quoting from things ... but I still wouldn’t want to start with this as the beginning of my literature review, because of how little the text is transformed.

Together, these observations may indicate that experts evaluate the quality of AI-generated literature reviews not only in terms of information synthesis, but also according to expectations of originality, meaningful textual reformulation, and sufficient variation from existing literature. However, these comments were identified only in isolated situations, which may suggest that concerns regarding limited textual transformation were not prominent across the generated outputs in the studied sample. LitTraceAI did not implement

direct plagiarism detection for its generated content. Nevertheless, the tool’s information transformation of the original sentences enabled Expert 4 and Expert 27 to recognise limited variation and similarities with the source material, suggesting that LitTraceAI may indirectly support the identification of potential plagiarism-related issues in generated content.

Relatedly, while analysing in detail the provenance content, Expert 26 referred to situations in which definitions should remain textually consistent because they represent established conceptual definitions within a research field:

...but if it is supposed to be a direct citation, it should be identical to the text in the cited paper. Since [LitTraceAI] opens quotation marks, I assume it is presenting a literal transcription from one or both sources, as these classical definitions tend to be reproduced across virtually all papers in the field. For that reason, it should not be paraphrased.

This observation introduces an additional perspective regarding paraphrasing in AI-generated literature reviews. In this case, the expert, whose research focuses on constructs such as Unethical Pro-Organisational Behaviour, suggested that certain definitions should remain stable because altering specific words through paraphrasing could change the conceptual meaning of the construct. This may be particularly relevant in research areas grounded in established theories, constructs, or formal definitions where textual consistency contributes to conceptual contextualization. Therefore, the expert’s comment may indicate that some forms of textual similarity are academically appropriate and even necessary, particularly when presenting exact definitions or direct quotations whose wording should remain unchanged.

One out of three experts emphasised additionally that plagiarism risks may also originate from the source materials themselves rather than exclusively from the generated output. Expert 1 explained:

The provenance should detect whether there is plagiarism from the authors of the papers it is citing because sometimes one replicates those mistakes. Usually, this happens when you are beginning your career as a researcher.

Although mentioned by a single expert, this perspective introduces a broader consideration regarding how AI systems may unintentionally propagate plagiarism problems already present in the original source material. The expert’s comment may indicate expectations that provenance mechanisms should not only support citation traceability, but also contribute to identifying potentially problematic content contained in the cited papers themselves, particularly for less experienced researchers who may be less equipped to critically assess and avoid reproducing such material. At the same time, this issue may require further research because LitTraceAI is grounded primarily in verifiability rather than factuality, meaning that the tool focuses on tracing generated sentences back to source content instead of evaluating whether the original source material itself contains plagiarism or other integrity related issues.

V.0.4 Constraints in Transparency

This theme reflects expert perceptions regarding opportunities to improve the transparency and interpretability of LitTraceAI. While experts acknowledged the presence of transparency mechanisms, they also referred to additional features that could help users better understand how generated sentences are supported and how the tool produces the final output. Importantly, the concerns raised by experts operated at different levels of relevance. Some observations were directly associated with the adequacy and justification of generated sentences, while others reflected preferences that may vary according to disciplinary background and research domain. Although the frequencies of some codes were low, the observations remain important because they introduce relevant considerations regarding trust, evidence sufficiency, and interpretability in AI-assisted literature review generation.

Two out of four experts referred to the lack of clear criteria regarding how many source sentences are required to support generated sentences. Expert 5 explained:

I would say that the tool [LitTraceAI] should include more source sentences to achieve greater precision, because when a generated sentence is supported by very few sentences, these small inaccuracies can arise. In other words, each generated sentence should be supported by more than one sentence, or by a sufficient number of sentences.

Similarly, another expert suggested:

An alert where the tool indicates a minimum number of sentences for each generated sentence in order to be sure about the response.

Additional concerns regarding citation support were identified during the experts' detailed analysis of the provenance information. Several experts questioned situations in which generated sentences were supported by only one sentence, particularly when the claims referred to broader conclusions or research consensus. Expert 22 explained:

For example, in provenance 31, identifying a gap based on a single reference is insufficient. One source alone makes the claim weakly supported; establishing a gap requires several references pointing toward a consensus. This is particularly problematic given that the relationship between customer experience and sustainability is noted as having been over- or underestimated.

Similarly, Expert 8 noted that in provenance 21 the original paper referred to “*some researchers*,” but LitTraceAI included only one citation:

*When the framework refers to **some researchers**..., it reproduces what the original paper states, supported by a single citation. However, since the claim invokes multiple researchers, the tool [LitTraceAI] should include more than one citation to adequately support it.*

Together, these observations may indicate that experts expect clearer indicators regarding what constitutes sufficient support for generated sentences. Rather than simply displaying supporting citations, experts appeared interested in understanding whether the quantity and representativeness of the supporting evidence are adequate for the type of claim being made. This concern appears particularly relevant because it relates to whether generated sentences can be considered sufficiently-justified rather than weakly-supported.

Relatedly, Expert 26 identified a different but complementary concern during the provenance analysis. While the expert acknowledged that some generated content was “*not false*” and that “*the definition of UPB is correct*” because LitTraceAI was taking the information “*textually*” from the source material, the expert also questioned generated statements that lacked sufficient supporting justification. Referring to the generated sentence “*Research has increased considerably,*” the expert explained:

Academic writing requires justification. The tool is drawing from someone else’s text, and that author did not justify the claim either.

While Experts 22 and 8 questioned situations in which one citation was insufficient to support broader claims, Expert 26 highlighted cases in which the generated sentence itself lacked supporting justification in the source material. In particular, the expert’s concern centred on the absence of a supporting citation for the claim “*Research has increased considerably,*” indicating that the generated sentence required explicit support through citation. Therefore, this observation may indicate that experts evaluate transparency not only in terms of the presence of provenance information, but also according to the sufficiency of the evidence supporting generated sentences.

One out of four experts additionally referred to the absence of quantitative confidence indicators associated with citation precision. Expert 11 suggested that “*the tool [LitTraceAI] itself should indicate the citation precision with a confidence level, similar to what is done in statistics.*”

In a follow up conversation, when asked why numerical confidence values were important given that LitTraceAI already provided support levels such as *Full Support*, *Partial Support*, and *No Support* together with provenance explanations, the expert explained that “*because of my engineering background and in my community quantitative metrics are desirable*”.

These observations may indicate that expectations regarding quantitative confidence indicators can depend on users’ disciplinary backgrounds and research domains. While some experts valued the existing transparency mechanisms that allowed them to understand how statements were generated and assess citation precision, as identified in Section S.0.1 of Appendix S, other experts may prefer additional quantitative indicators to support interpretation and evaluation of the generated sentences. Compared to the concerns regarding insufficient or unjustified evidence, this observation appears more dependent on user background and disciplinary expectations.

Finally, one out of four experts referred to the absence of a visible overview explaining how LitTraceAI generated the final output. Expert 25 explained:

There are no messages showing how the workflow occurred. The screen displays the sections directly, but [LitTraceAI] does not explain how it got there.

The observation of Expert 25 is particularly relevant given the expert’s background in technology adoption research. In a follow up conversation, the expert explained that “*in my research and outreach experience when tools are based on complex information like is the case of the literature reviews, users want to understand the workflow of the tool.*” The expert further noted that, particularly in the current AI boom, “*users need to see from beginning to end the workflow,*” even if they do not fully understand the technical details, because visualising the process can foster trust in the tool “*to some extent,*” depending on the quality of the generated output.

Although the workflow described in Section 4.2 was considered during the design of LitTraceAI, the study primarily evaluated whether a workflow grounded in the rhetorical roles

of sentences improved the quality of generated literature reviews beyond simple summarisation techniques. Additionally, the evaluation sessions lasted approximately two hours on average, which introduced practical time constraints regarding the implementation and evaluation of additional workflow visualisation features. Nevertheless, this functionality could be implemented relatively easily in future research. Similar to the interaction features discussed in Section V.0.1, such as direct access to the original PDFs and highlighting the source passages used to generate statements, visualising the workflow from beginning to end would foster user trust in AI-assisted literature review systems.